Community Science: Prediction to Arrival, Modeling H5N1 in New Zealand

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***Abstract*—Highly Pathogenic Avian Influenza (HPAI) H5N1 reached Aotearoa New Zealand in 2026, creating an urgent need to understand its likely impact on vulnerable bird populations before comprehensive monitoring data becomes available. Building on earlier work that correctly identified Brown Skua as H5N1's likely vector into New Zealand, we extend this community science approach to two terrestrial raptors, Kāhu (Swamp Harrier) and Kārearea (New Zealand Falcon), using eBird checklist frequency as a proxy for population decline. For each species we fit a reference/control species pair drawn from a documented overseas H5N1 outbreak, using a control species in a different ecological niche to attribute any decline specifically to the outbreak rather than observer effort or seasonal variation. We validate this approach against the Great Skua, where eBird's own decline estimate (42.2%) tracks an independently field-measured population decline (66%), then apply the same method to project Kāhu and Kārearea's checklist-frequency trajectories over the next two years in NZ. This paper demonstrates one use case of how large-scale community science datasets can be repurposed for rapid, defensible modeling when conventional monitoring cannot keep pace with an emerging outbreak.**

***Index Terms*—Animals, Data processing, Environmental factors, Epidemiology, Modeling, Statistics**

I. Introduction

The widespread adoption and growth of citizen or community science has resulted in a flood of accessible, large-scale, crowd-sourced observational datasets – with projects in physics, life science, medicine, ecology, biology, and many other fields [1]. Some of these platforms are massive. For instance, eBird [2], [3] contains over a billion observational records [4], with over a million users and 100 m checklists [5], and iNaturalist [6] has almost 250 m observations, with over 3 m observers for over 50 k species [7].

While these large datasets offer immense research potential, the inherent heterogeneity of structure, data quality, curation standards and procedures presents significant challenges that limit their impact as research data. Specifically, community science datasets typically consist of observations of differing quality, from observers of differing skill, using differing protocols [8]. Nevertheless, these datasets as they currently stand are used to produce scientific research and as a consequence to inform public policy globally.

One use of these uses is to inform epidemiological models. The H5N1 strain of highly pathogenic avian influenza (HPAI) has decimated bird populations globally. It has caused unprecedented mass mortality events in South America [9], [10], North America [11], and Europe [12], [13]. HPAI has recently spread through the subantarctic [14]–[16] and to Antarctica [17]. In the past couple of months, the first few records have arrived in Australia [18], with 74 confirmed records to date, and in Aotearoa New Zealand (NZ) [19], with two confirmed records. Observing, collecting, and testing birds for HPAI is costly and difficult, and limited by resources and funding [20]. In this case, public datasets made by volunteer community scientists can be useful for showing trends in bird populations [21], [22]. This has precedence, as some studies of HPAI have already used community science datasets to look at spillover effects on poultry [23].

In [24], we showed that combining published metrics of bird mortalities from HPAI in the subantarctic with community science data from there could be used to infer future population metrics for birds in NZ. We harmonized community science data from eBird, iNaturalist, and other datasets to build integrated species distributional models (SDMs), an approach taken by others [25]–[27]. In that paper, we identified Brown Skua *Stercorarius antarcticus* as one of the likely vectors for HPAI into Australasia, following others [28], [29]. Since publication, one of the confirmed cases in NZ is a Brown Skua, with the other being a Kāhu Swamp Harrier *Circus approximans* [19].

Skuas kleptoparasitize and predate other birds, either directly or as carrion, increasing their exposure to HPAI. HPAI greatly affects skuas, and has reduced the breeding population of Great Skua *Stercorarius skua* in Scotland by 66% [12], [30], [31]. NZ's bird populations are already vulnerable; a third of native bird species have become extinct since colonization [32]. New Zealand has many rare endemic species, of which the five species of Kiwi (from the endemic order Apterygiformes), Takahē *Porphyrio hochstetteri*, and Kākāpō *Strigops habroptilus* are some of the most charismatic. For birds of prey, there are only two diurnal terrestrial species, not including the skua: the Kāhu and the Kārearea. The Kārearea New Zealand Falcon *Falco novaeseelandiae* is the only falcon in NZ, is an endemic species, and is considered Threatened [33]. The Brown Skua is also considered Nationally Vulnerable [33], a more dire status.

Drawing on our previous work, and noting the two affected species so far, we here present updated models of bird population declines for the the three main species of predatory birds in NZ: Brown Skua, Kāhu, and Kārearea New Zealand Falcon *Falco novaeseelandiae*. We base our models on population declines that are present in recorded eBird data from the northern hemisphere.

Community science platforms rely upon experts to review and curate the data provided by volunteers. For eBird [2], the quality control system flags unusual observations for review by volunteer reviewers. These systems work at scale, but they are prone to certain biases. For birds, some of these issues are well known [34], [35]: sampling bias, observer skill variation, detection bias, duplicated observations, over-reliance on AI tools like [36], and gamification [37], [38]. It is generally up to the researcher to control for these biases. Community science platforms allow their data to be downloaded and used, often for free and with permissive data licenses.

Tapping into this immense reservoir of data, there is nascent field validating statistical models run on community science datasets (c.f. [35], [39]–[41]), often, urging caution when applying models, as the data can be dirty. [42]. Yet community science data has been used to study bird mortalities [43]: one study tracked mortality rates for gannets *Morus bassanus* after HPAI with iNaturalist, finding that these datasets can track the species, timing, and location, but not breadth or mechanism, of epidemiological events [44]; another showed community science data had a strong relationship to rigorous academic studies, when showing Peregrine Falcon *Falco peregrinus* declines [45].

In this paper, we discuss our methods of using reference species, combined with control species with similar distributions and habitats, to build a model of population declines given decreases in observational data. We built box plots that can be used to quickly visualize temporal differences, as well as generalized linear models (GLMs) to show more robust population curves. We project population declines for NZ birds, assuming an initial HPAI wave begins in the next few months. We also pin these models to published mortalities, à la [24], to mitigate community science biases and ensure more robust data.

II. Methods

A. Choosing species

We selected a small set of target species for projection statistics: Brown Skua, Kāhu, and Kārearea.[1] The first two were chosen as the only two species with positive detections of H5N1 in New Zealand to date [56], while the Kārearea was added as the other extant bird of prey in New Zealand, not including three extant species of owls [57]. Raptors are also known reservoirs and indicator species for HPAI H5N1 [52].

The criteria for reference species inclusion was: a) genetic similarity to the New Zealand target species b) plentiful data within [5] c) similar ecological niches and behavioral patterns, and d) published reports of HPAI infection. Candidates were hand-chosen and not graded on these criteria. For each species, a control species was chosen with similar habitat but different ecological, behavioral, and genetic make-up, and thus with different profiles for H5N1 infection. These control species were found on similar observational checklists as the target species, which controlled for some (but not all) differences in logging patterns: across all pairs, 17.8–51.7% of a target species' checklists also reported its control. The chosen species are presented in Tab. I.

For Skua, the reference species was Great Skua, as it is within the same genus and its decline has been well-documented in Scotland. For Kāhu, the reference species is a composite of Common Buzzard in Germany and Rough-legged Hawk in the US. For Kārearea, Peregrine Falcon was chosen as a congener; however, hunting patterns may differ between the two (see Sec. IV).

B. Data collection

For each chosen taxon, data was downloaded from eBird [58] using their online web platform. The downloaded data was for ten years of records (from 2016), in the regions provided in Table I, including sampling data which includes abundance data relative to other species. A CSV for all files with the following fields has been uploaded to Zenodo [59].

We filtered the data, limiting the result to deduplicated complete checklists. On eBird, "complete" checklists control for effort by containing counts for all identified species. We generated controlled time series by binning the number of checklists with target species detected against all filtered checklists in that region, per week.

C. Modeling

We performed a literature search for published reports of cases of HPAI in these species in the target areas, focusing on the first available published date of HPAI incursion. While subsequent waves of infection were noted, they were not used in the generated box plots, which are complementary to the GLMs, and could be represented as:

$$\tilde{x}_{s,w} = x_{s,w}/b_s \tag{1}$$

where $x_{s,w}$ is species s's checklist frequency in week w and b_s its mean pre-outbreak frequency (season-restricted for strongly seasonal species). Boxes plot the 2.5th/25th/50th/75th/97.5th percentiles of $\tilde{x}_{s,w}$ per species-year, in place of the standard $1.5\times$IQR whisker rule, with a line connecting each species' per-year median.

We limited GLMs to individual reference–control pairs rather than pooling data across species or regions, thereby curtailing potential confounding. We selected a negative binomial structure because a Poisson GLM was invalidated by overdispersion in the weekly bins. Following [60], seasonal variation was controlled using a four-parameter annual and semiannual harmonic (sin/cos) fit. Finally, we isolated the deseasonalised trend by evaluating the model with harmonics fixed at zero,

[1]We use Māori names where they are used frequently in New Zealand English. Hākoakoa is one of the Māori names for Brown Skua.

Region	Target and Reference Species	Control species	Documented HPAI outbreaks	References
NL	Peregrine Falcon *Falco peregrinus*	European Robin *Erithacus rubecula*	2017, 2022, 2023	[46]–[49]
US	Peregrine Falcon	American Robin *Turdus migratorius*	2022	[50]
DE	Common Buzzard *Buteo buteo*	European Robin	2020	[51], [52]
US	Rough-legged Hawk *B. lagopus*	Horned Lark *Eremophila alpestris*	2021, 2022	[53]
UK	Great Skua *Stercorarius skua*	Meadow Pipit *Anthus pratensis*	2021, 2022	[54], [55]
NZ	Kārearea *F. novaeseelandiae*	Silvereye *Zosterops lateralis*	None	–
NZ	Kāhu *C. approximans*	Australian Magpie *Gymnorhina tibicen*	2026-07-17 (n=1)	[19]
NZ	Brown Skua *S. antarcticus*	New Zealand Pipit *A. novaeseelandiae*	2026-07-17 (n=1)	[19]

TABLE I
TARGET, REFERENCE AND CONTROL SPECIES FOR SDMS.

determining relative decline through a comparison of the pre- and post-outbreak trend means.

The reference and control species pairs are fitted with the full model above; NZ target and control species receive a single-species trend fit restricted to pre-outbreak data only, up to the last available datum (29 June 2026). The equation for spline (using patsy v1.0.2 [61]) and seasonality modeling is given below:

$$\begin{aligned} n_t \; &\sim \mathrm{NegBin}(\lambda_t, k) \\ \log(\lambda_t) \; &= f(t) \; + \; I_{\text{ref}}\, f_{\text{ref}}(t) \; + \; f_{\text{season}}(t) \; + \; \log(E_t) \end{aligned} \tag{2}$$

Here n_t is the weekly checklist count; λ_t its expected value under a negative binomial distribution with dispersion k; $f(t)$ a smooth (5-degree-of-freedom natural cubic spline) trend shared with the control species; $f_{\text{ref}}(t)$ the reference species' own smooth deviation from that trend, switched on only for reference-species rows via the indicator I_{ref}; $f_{\text{season}}(t)$ a smooth (harmonic) seasonal term; and E_t the week's total checklists, entered as an offset to normalize for observer effort.

Once GLM curves are computed for reference and control species, the post-outbreak shape (per reference and control species) is re-indexed to weeks since outbreak start, and then anchored onto the target species timeline, assuming an HPAI incursion on 29 June 2026 (chosen for data availability, and not assuming the two cases represent the beginning of a full outbreak). Both curves are normalized to pre-outbreak baselines, and given a 2-year (112 week) horizon. This GLM can be expressed like Eq. 2 but without $I_{\text{ref}} f_{\text{ref}}(t)$, since each species is fit on its own rather than jointly, without post-outbreak NZ data.

Some target species curves are computed from a multi-species reference envelope: kārearea (NL and US Peregrine Falcon) and kāhu (DE Common Buzzard and US Rough-legged Hawk) each draw on two reference species. This is reported as min/max envelope, and not blended or averaged.

Each reference and control pair is then classified by a likelihood-ratio test comparing the model above against a reduced model with the reference-deviation term dropped, testing whether the reference species' fitted curve differs in shape from its control's. A pair is corroborating if this difference is significant ($\alpha = 0.05$) and the reference's deseasonalized relative decline is at least twice the control's (or the control does not decline).

TABLE II
NEGATIVE-BINOMIAL GLM PER REFERENCE/CONTROL PAIR: A LIKELIHOOD-RATIO TEST FOR WHETHER THE REFERENCE'S FITTED DECLINE DIFFERS FROM ITS CONTROL'S, THE SAME COMPARISON SHOWN VISUALLY FOR ONE PAIR IN FIG. 1. THE DIFFERENCE IS SIGNIFICANT ($p < 0.001$) FOR ALL PAIRS, AND EACH REFERENCE'S DECLINE IS AT LEAST $2\times$ THE CONTROL'S (OR THE CONTROL DOES NOT DECLINE).

Region	Reference (Control)	Declines
NL	Peregrine Falcon (E. Robin)	28.0% (5.6%)
DE	Common Buzzard (E. Robin)	7.1% (-9.1%)
US	Peregrine Falcon (A. Robin)	35.1% (-4.4%)
US	Rough-legged Hawk (H. Lark)	48.6% (22.8%)
GB	Great Skua (M. Pipit)	42.2% (-9.2%)

Python was used for filtering and the GLMs, as well as for table generation, while R was used for generation box plot figures.

III. RESULTS

GLM curves were tested for goodness-of-fit; McFadden's pseudo-R^2 diagnostics for all five fits were within acceptable parameters (0.06-0.27, with NL Peregrine Falcon being the most stable). Pseudo-R^2 reflects overall model fit, not the specific reference-control divergence the verdict is based on.

The GLM results for reference species, shown in Tab. II, show a significant decline compared to control species; the rate of observation suggests population declines, for which HPAI is the most likely factor given epidemiological studies. Great Skua is the most salient number, as a 42.2% decline in observations is similar to a known mortality rate of 66% [12], [30], [31].

Brown Skua's projected checklist frequency is expected to fall 45.5% by two years post-outbreak (Based on the UK Great Skua decline). Kārearea's projected checklist frequency is expected to fall 30.2% to 33.7% by two years post-outbreak (NL and US Peregrine Falcon declines). Kāhu's projected checklist frequency is expected to fall anywhere from 2.3% to 52.0% by two years post-outbreak (DE Common Buzzard and US Rough-legged Hawk).

The box plots based on Eq. 1 are displayed in Fig. 1; the UK GLM curves are displayed in Fig. 2. Limitations are discussed below. A NZ projection for Brown Skua is shown in Fig. 3. Projections for Kāhu and Kārearea are given in Figs. 4 and 5.

IV. DISCUSSION

The Great Skua–Brown Skua analogue is relatively straightforward: both are pelagic with similar ecological niches,

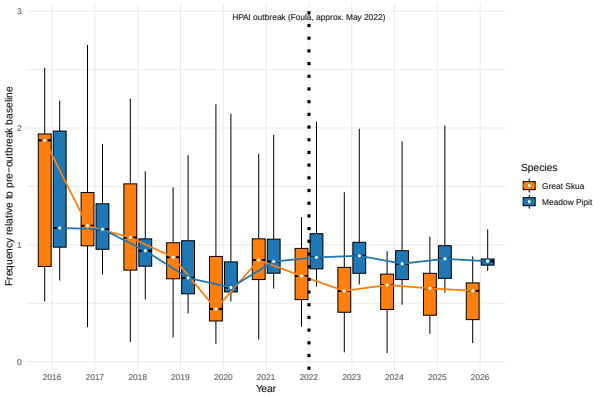


Fig. 1. H5N1 impact on Great Skua against the reference Meadow Pipit population, by year (2.5th–97.5th percentile boxes, relative to each species' own pre-outbreak summer baseline).

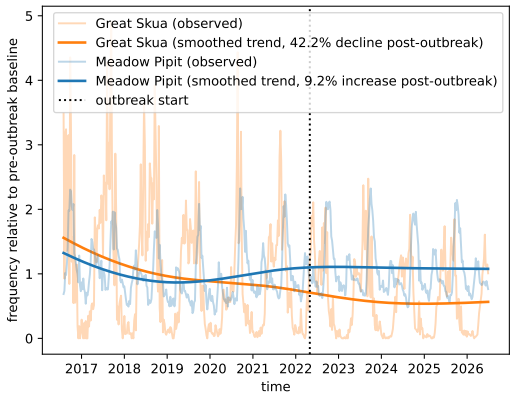


Fig. 2. Deseasonalized fitted trend (negative-binomial GLM, Eq. 2) for Great Skua and Meadow Pipit, same pair as Fig. 1, raw weekly points faded behind the fitted curve.

though Great Skua breeds locally in Scotland whereas Brown Skua rarely breeds near humans in NZ. The Kāhu–Common Buzzard/Rough-legged Hawk comparison is more of a stretch: both reference species are well represented in eBird, but habitats differ somewhat and neither shares Kāhu's genus. The Kārearea–Peregrine Falcon comparison should likewise be tempered: Peregrine Falcon preys on waterbirds far more often, and a kestrel or sparrowhawk may have been the closer ecological fit.

Our primary goal was to compare well-documented HPAI statistics across species, showing how projections can be extended from one taxon to another using community science data. eBird showed Skua declines comparable to published metrics, and the resulting projections are similar in scope to those we anticipated for NZ's subantarctic islands in [24]. Kāhu's large envelope shows these projections can vary widely; Kārearea's smaller envelope suggests a single control species may sometimes be sufficiently robust. These projections are not forecasts; we deliberately avoid citing mortality metrics, since a more complex integrated GLM or

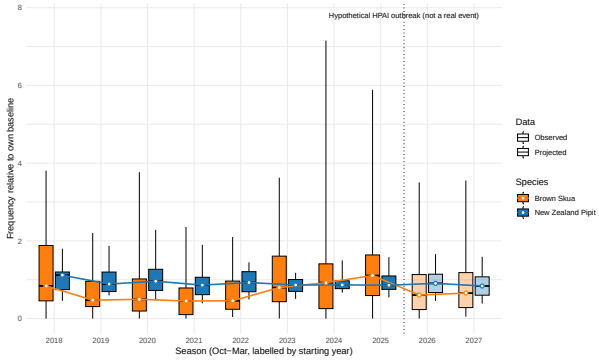


Fig. 3. A projected trajectory for Brown Skua based on Great Skua in the UK.

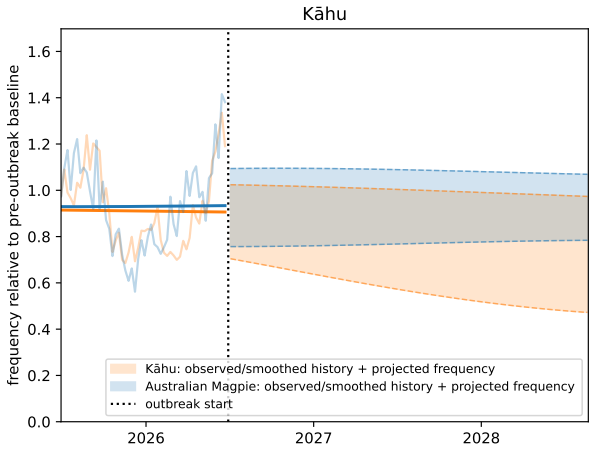


Fig. 4. A projected trajectory GLM for Kāhu, with an envelope derived from DE Common Buzzard and US Rough-legged Hawk.

SEIR model would be needed for that purpose. The GLM has limitations: a control species rules out reporting-behavior artifacts but not other real-world causes, such as weather, which may affect skuas differently from HPAI (though see [55]); and acute HPAI mortality risks being smeared across weekly bins, particularly for community science protocols lacking standardized reporting periods.

A novel element of this paper is the use of a control species occupying a different ecological niche but similar habitat and observational frequency in order to isolate HPAI incursion as the driver of decline. Raptor frequency drops could equally reflect behavioral shifts, climate change, or changes in observation effort during COVID or by season; a co-occurring control species in a different niche helps limit these confounds.

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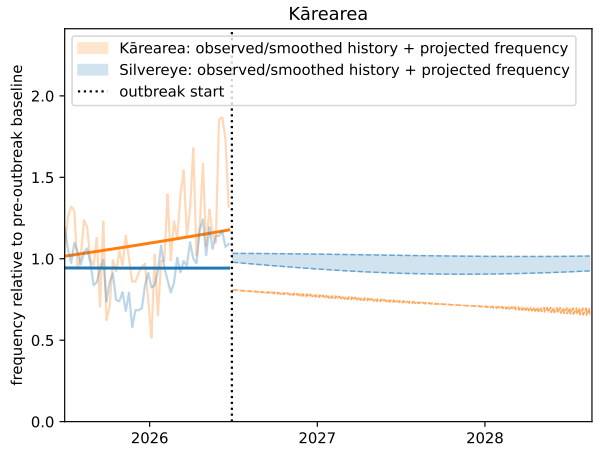


Fig. 5. A projected trajectory GLM for Kārearea based on Peregrine Falcons in the US and the NL.

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